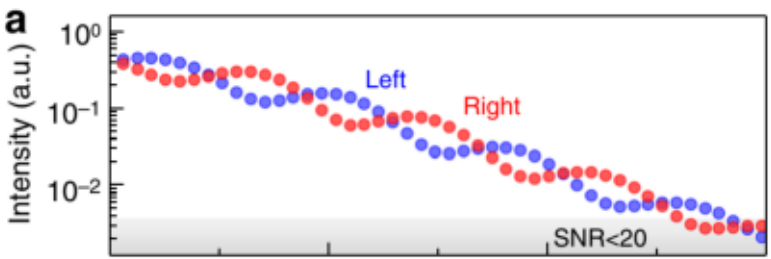
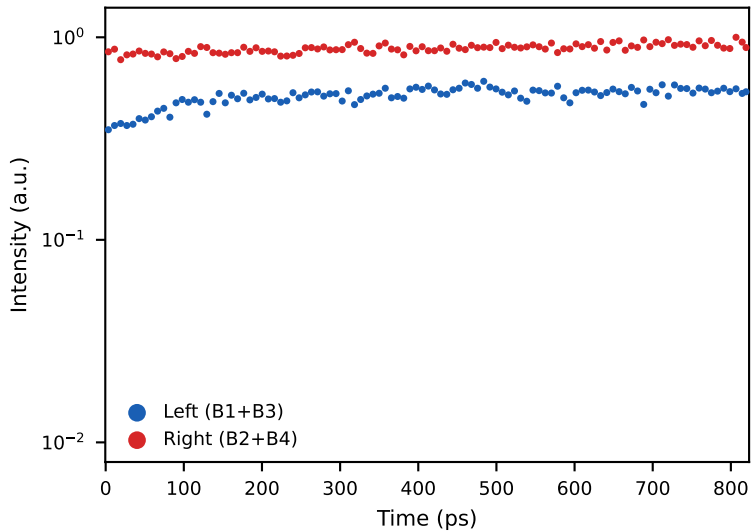
Adiyatullin et al. 2017 (Nat. Commun. 8:1329), Fig. 2(a)



This work: 26-08-04 HBT, 20000 frames, CFD50-timed



Left: original Fig. 2(a) of Adiyatullin et al. 2017 (Nat. Commun. 8:1329)— accumulated emission intensity of the left (blue) and right (red) mesa on a logarithmic scale, showing anti-phase population oscillations; the gray band marks the low-confidence SNR<20 region. Right: this work — replication on the 26-08-04 HBT batch (20000 single-sweep frames; photon-event source, rebin 8≈ 7.9 ps/bin, peak-normalized). Each mesa is recorded simultaneously by both HBT arms; the plotted intensity sums its two arm bands (Left = B1+B3, Right = B2+B4). The per-frame time zero is calibrated by CFD50, leading-edge constant-fraction discrimination on the reference pulse (50%-of-peak crossing, sub-row interpolated), placing the origin ~2.7 ps earlier than the former half-max centroid definition; the measured StreakCamBackG background is subtracted. Outcome: no anti-phase oscillation— cross-arm z fluctuations are uncorrelated (r=0.03–0.10, p>0.14); the 95% upper limit on a phase-locked sinusoidal z amplitude is 0.012–0.019 for 3–200 ps periods (~1–2% intensity modulation); the t≤150 ps left-side dip is an overdamped monotonic recovery, not an oscillation.